ALGEBRA 1 · LINEAR FUNCTIONS AND GRAPHING

Slope-Intercept Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Slope-Intercept Explorer** on Mr. K's Classroom: Algebra 1 → Linear Functions and Graphing → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. In y = mx + b, what do you think happens to the line when b gets bigger?

2. What do you think a line with a negative slope looks like?

2 Explore use the tool and record what you see

Change b (keep rise 2, run 1, so m = 2)

b	Equation	y-intercept	Compared with y = 2x
−3			
0			
2			
5			

Change the slope (keep b = 1)

Rise	Run	Slope m	Uphill, downhill or flat?	Steeper than y = x?
1	2			
3	1			
−2	1			
−1	3			
0	4			

Compare with

Turn on the parallel and perpendicular lines.

Your slope	Parallel slope	Perpendicular slope
$\frac{2}{3}$		
−4		
1		

3 Write the rule fill in each blank from what you observed

1. Changing b moves the line _____ or _____ without changing how steep it is.
2. When b is negative, the line crosses the y-axis _____ the origin.
3. When m is positive the line goes _____ from left to right. When m is negative it goes _____.
4. The larger the absolute value of m, the _____ the line.
5. Parallel lines have _____ slopes. Perpendicular slopes are opposite _____.

4 Check no tool: show your work

1. Give the slope and y-intercept of $y = -3x + 4$.

2. Write the equation of the line with slope $\frac{1}{2}$ and y-intercept $(0, -2)$.

3. Write the line parallel to $y = 5x - 1$ that crosses the y-axis at 3.

4. What slope is perpendicular to $y = -\frac{2}{5}x + 7$?

5 Reflect

Explain what the slope tells you, using the words rise and run.